# Modeling U.S. Tornado Damage

Luke O'Neill, Averey Copeland, Dale Osborn
University of Colorado - Boulder

Applied Mathematics
COLLEGE OF ARTS AND SCIENCES

## Introduction

The National Oceanic and Atmospheric Administration or NOAA has maintained a nationwide database of all major weather events that have occurred since January 1950. Tornadoes are among the most destructive and unpredictable of these events, causing erratic amounts of damage at a moment's notice. Sometimes an event's total damage cost cannot be counted for until well after an event has occurred, potentially delaying vital relief efforts. By looking at information that can be gathered quickly, such as event duration, F-class, and location, we can predict the total damage caused by an event and assist emergency management agencies, governments, and local communities in allocating resources more effectively before and after tornado events.

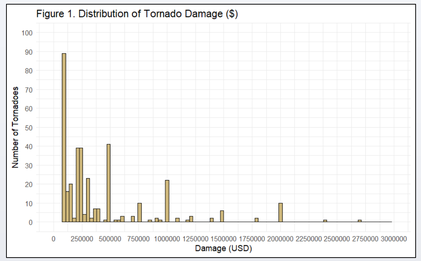
## Methods

We obtained our tornado data from NOAA's storm event database. NOAA is a national center for environmental information that has been recording environmental data for the last 200 years

To prepare the data we filtered the storm event type for only tornadoes, then we combined all direct and indirect injuries and deaths into one variable. We also calculated the range of the tornado by subtracting the tornado's distance from an unspecified city, town or village at its start and end and taking the absolute value. We also had to convert the damage into numerical data type as it was listed in thousands and millions of dollars. Finally we selected all of the variables that are used for tornadoes. Our dataset has 1764 observations where each observation is a tornado that is described by 14 variables 10 of which are numerical and 4 of which are categorical.

| Variable | Units | Min | Median | Max |
|---|---|---|---|---|
| Property Damage | $ | 0 | 5000 | 475,000,000 |
| Range | Miles | 0 | 1 | 12 |
| Tornado's path length | Miles | .01 | 2.24 | 49.96 |
| Tornado width | Yards | 1 | 100 | 3200 |
| Injuries and Deaths | Persons affected | 0 | 0 | 87 |

## Exploratory Results



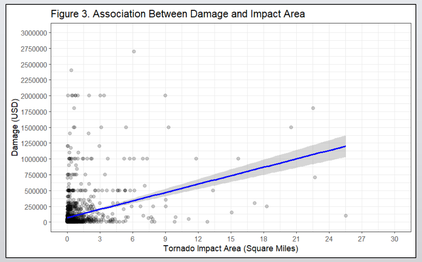
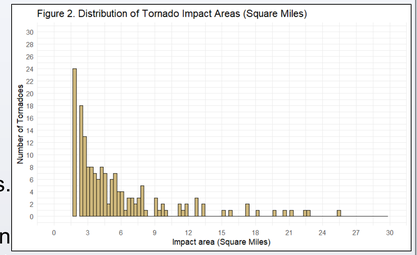
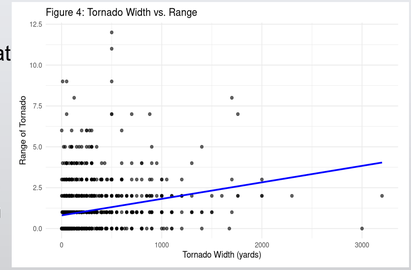
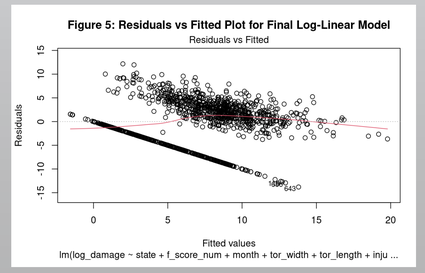



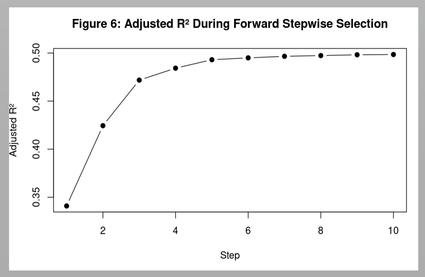

**Fig. 1** shows the distribution of tornado damage. The distribution is heavily right skewed, where most tornadoes don't cause much damage, with a lesser amount reaching up to 92 million dollars in damages. This showed modeling damage directly would cause issues. **Fig. 2** shows the distribution of the tornado impact area. The distribution is heavily right skewed, implying that smaller tornados are much more common. **Fig. 3** shows that the relationship between impact area and damage is positive but weak with heteroskedasticity. Showing that the spread of damage increases with impact size. Most data is clustered near 0 and causing little damage. Indicating the impact area is not a good predictor alone. **Fig. 4** shows that the relationship between width and range is positive and linear. Telling us that wider tornadoes will travel further, which makes sense intuitively.



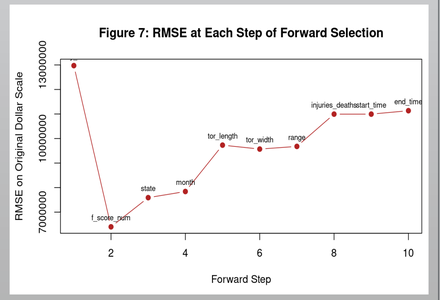



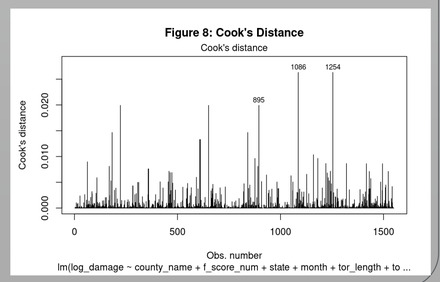

## Predictive Results




We selected a multiple linear regression model, which has the assumptions of linearity, constant variance, and normal residuals. **Fig 5** shows these are not met perfectly, and that many tornados cause 0 damage while our model predicts positive damage. The log transformation helped stabilize the variance and reduce skew. We used forward stepwise selection to pick our predictors, and the R-squared for each step is shown in **Fig 6.** Due to the unpredictability of tornado data, the RMSE for each step(shown in **Fig. 7**) are all large. **Fig 8.** shows the influence each observation has on our model, indicating that a few tornadoes have a strong influence on our linear model




## Discussion

Our model significantly outperforms the null with a residual standard error about 30% better than the null model. Our model is also able to explain about 50% of the variance in damage cost. The ANOVA P-Value indicates that our model is very statistically significant and the chance our model is performing this well by chance is very low.

| Metric | Null Model | Fitted Model |
|---|---|---|
| Adjusted R^2 | 0 | .4984 |
| Residual Standard Error | 5.572 | 3.946 |
| ANOVA P-value | NA | 2.2e^-16 |

We decided to use the log linear regression model to help account for the heavily right-skewed data clustered near 0. A technical challenge of the log linear regression model is the assumption of linearity and constant variance limiting our models ability to represent tornado damage. Practically, our model is significant as it could be used to help communities prepare for events they would not have otherwise anticipated. Contributing to fewer casualties in a tornado and the protection of property that would have otherwise been overlooked. Ethically, this could be problematic as people could overly rely on the imperfect model. Either leading to people being underprepared or causing unnecessary panic in a community. Also, since the model considers county as a predictor variable it may cause lower income counties to have underestimated risk of damage.

## Conclusions

After modeling tornado damage using a log-linear regression model and using forward stepwise selection, we found a model with an adjusted R-squared value of 0.498. This means that the model explained 49.8 percent of the variability in tornado damage, which is a good result given the heavily right-skewed data. Future work could improve the model by incorporating more predictors. Predictors such as population density and average building age would provide more context to the model and strengthen our findings. A more accurate and complete version of this model would provide a more reliable model for weather services and emergency services.